

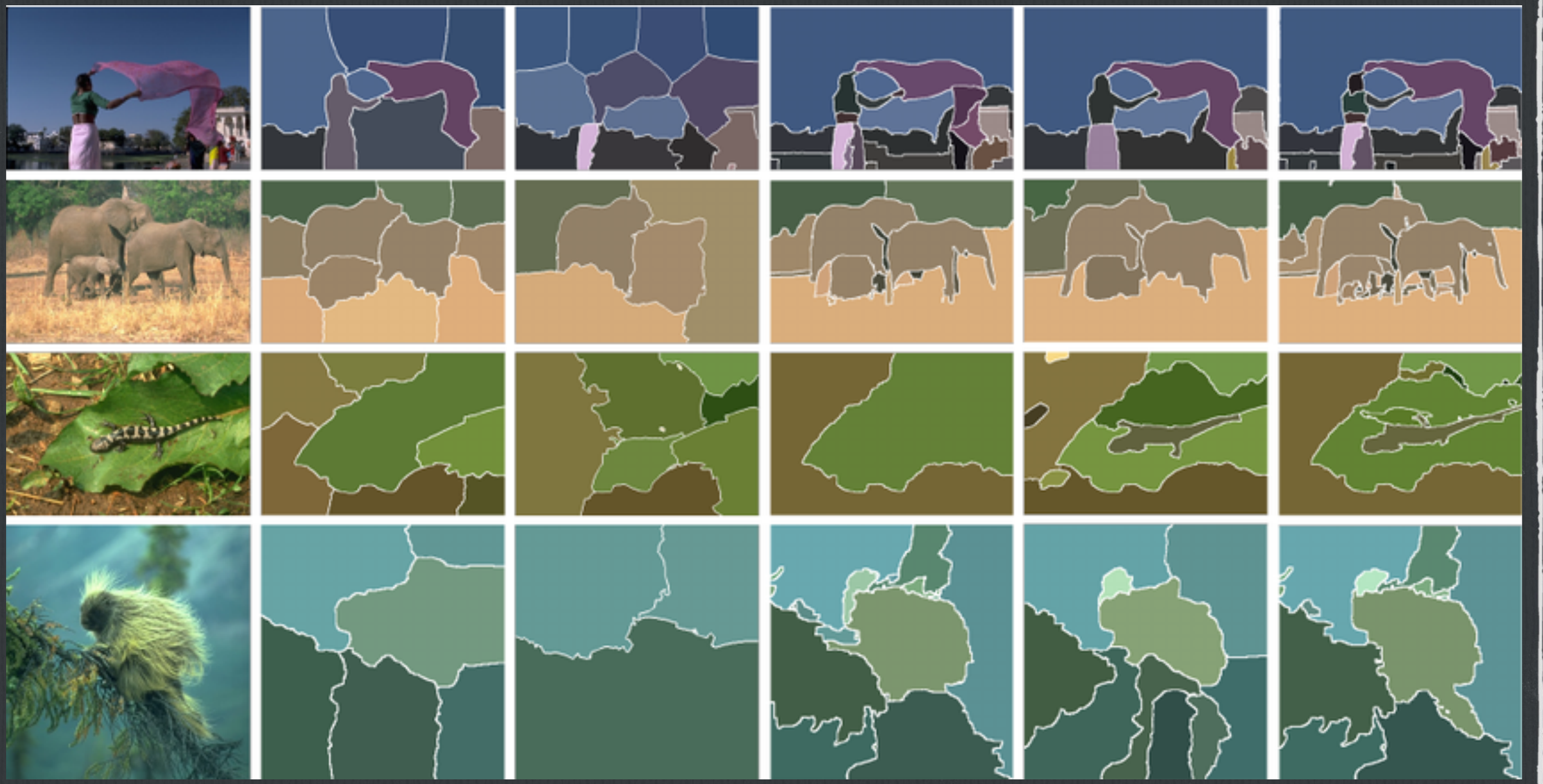
Digital Image Processing

Chapter 6: Edge Detection

Introduction

Preview

- Segmentation is to subdivide an image into its component regions or objects.
- Segmentation should stop when the objects of interest in an application have been isolated.



Principle Approaches

- Segmentation algorithms generally are based on one of 2 basis properties of intensity values:
  - **discontinuity** : to partition an image based on sharp changes in intensity
  - **similarity** : to partition an image into regions that are similar according to a set of predefined criteria.

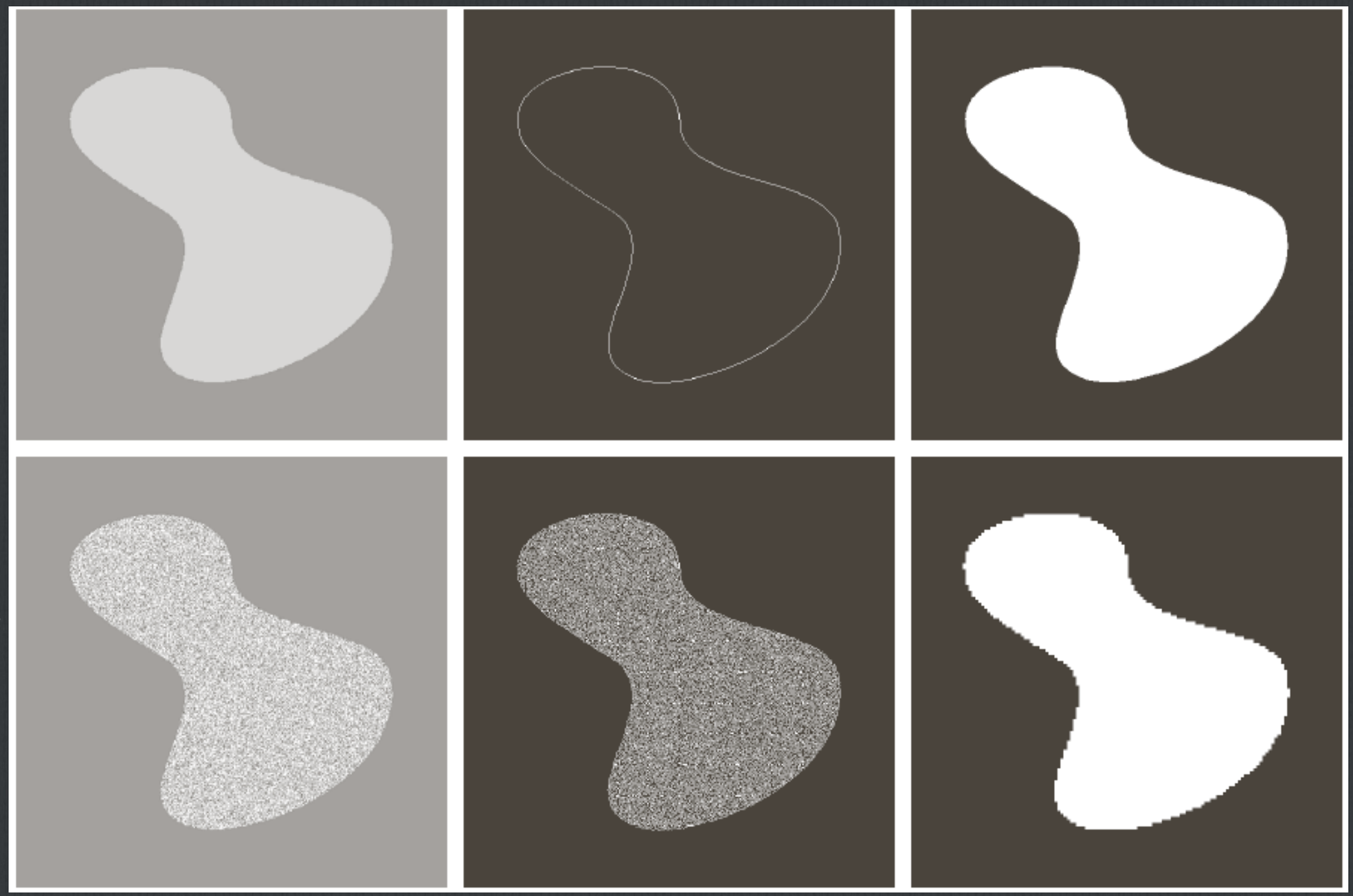
Fundamentals

A more formal definition – Let R represent the entire image.

Segmentation is a process that divides R into n subregions $R_1, R_2, ..., R_n$ such that:

1. $\bigcup_{i=1}^{n} R_i = R$.
2. R_i is a connected set for each $i = 1, 2, \ldots, n$.
3. $R_i \cap R_j = \emptyset$ for all i and j, $j \neq i$.
4. $Q(R_i) = TRUE$ for each $i = 1, 2, \ldots, n$.
5. $Q(R_i \cup R_j) = FALSE$ for any adjacent regions R_i and R_j.

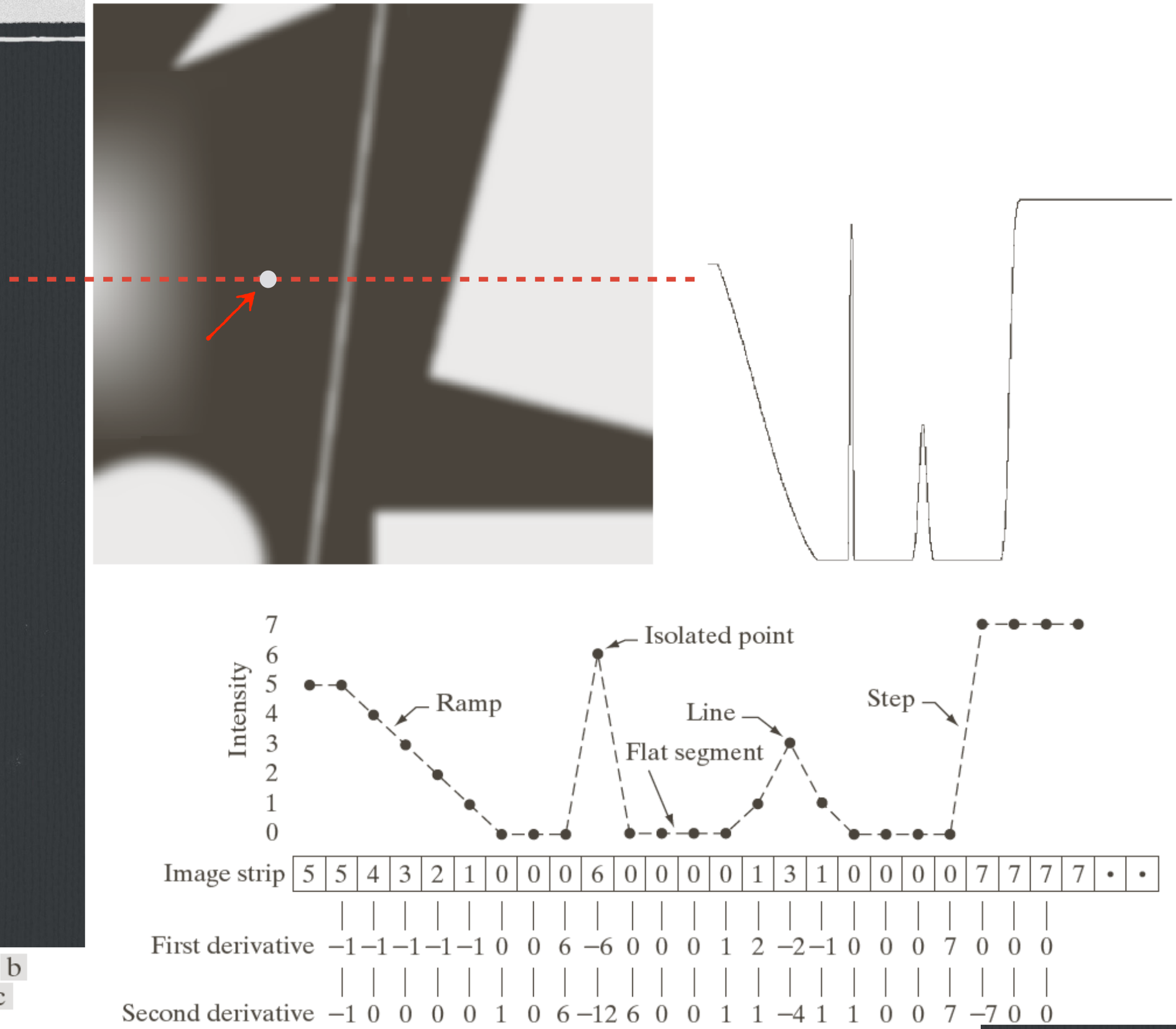
Here $Q(R_k)$ is a predicate that indicates some property over the region.



a b c
d e f

FIGURE 10.1 (a) Image containing a region of constant intensity. (b) Image showing the boundary of the inner region, obtained from intensity discontinuities. (c) Result of segmenting the image into two regions. (d) Image containing a textured region. (e) Result of edge computations. Note the large number of small edges that are connected to the original boundary, making it difficult to find a unique boundary using only edge information. (f) Result of segmentation based on region properties.

Mathematical Background



a b
c

FIGURE 10.2 (a) Image. (b) Horizontal intensity profile through the center of the image, including the isolated noise point. (c) Simplified profile (the points are joined by dashes for clarity). The image strip corresponds to the intensity profile, and the numbers in the boxes are the intensity values of the dots shown in the profile. The derivatives were obtained using Eqs. (10.2-1) and (10.2-2).

Derivatives

- We will use derivatives (first and second) in order to discover discontinuities.

- If we observe a line within an image we can consider it as a one-dimensional function f(x). We will now define the first derivative simply as the difference between two adjacent pixels.

$$\frac{\partial f(x)}{\partial x} = f'(x) = f(x+1) - f(x)$$

$$\frac{\partial^2 f(x)}{\partial x^2} = f''(x) = f(x+1) - 2f(x) + f(x-1)$$

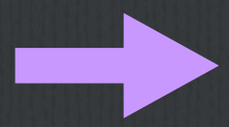
Derivatives Properties

- ☐ First derivative generally produce thicker edges in an image
- ☐ Second derivative has a very strong response to fine details and noise
- ☐ Second derivative sign can be used to determine transition direction.

Derivatives - Computation

- The derivative is a linear operator ant therefore can be computed using a mask
- For example, the mask for the second derivative on the x direction would be:

0	0	0
-1	2	-1
0	0	0



w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

FIGURE 10.3
A general 3×3 spatial filter mask.

Detection of Isolated Points

Detection of Isolated Points

- Based on the fact that a second order derivative is very sensitive to sudden changes we will use it to detect an isolated point.

- We will use a Laplacian which is the second order derivative over a two dimensional function.

Laplacian

$$\nabla^2 f(x,y) = \frac{\partial^2 f(x,y)}{\partial x^2} + \frac{\partial^2 f(x,y)}{\partial y^2}$$

$$\frac{\partial^2 f(x,y)}{\partial x^2} = f(x+1,y) - 2f(x,y) + f(x-1,y)$$ **x-direction**

$$\frac{\partial^2 f(x,y)}{\partial y^2} = f(x,y+1) - 2f(x,y) + f(x,y-1)$$ **y-direction**

$$\nabla^2 f(x,y) = f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1) - 4f(x,y)$$

- The mask for the Laplacian is:

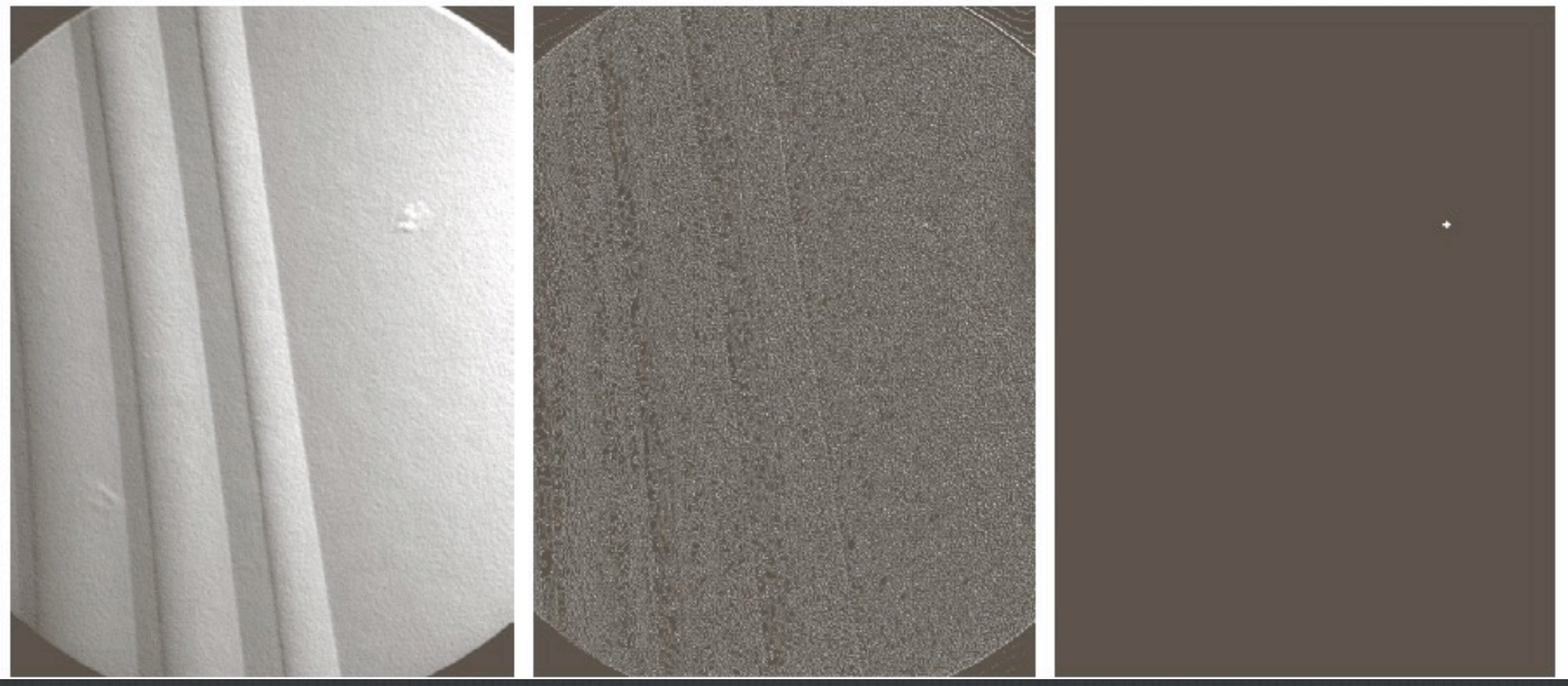
0	1	0
1	−4	1
0	1	0

1	1	1
1	−8	1
1	1	1

- We will now decide about a threshold T, and any pixel whose value is larger than T after being masked will be considered as an isolated point.

$$g(x,y)=\begin{cases}1 & \text{if } |Z(x,y)|>T\\ 0 & \text{otherwise}\end{cases}$$

1	1	1
1	−8	1
1	1	1



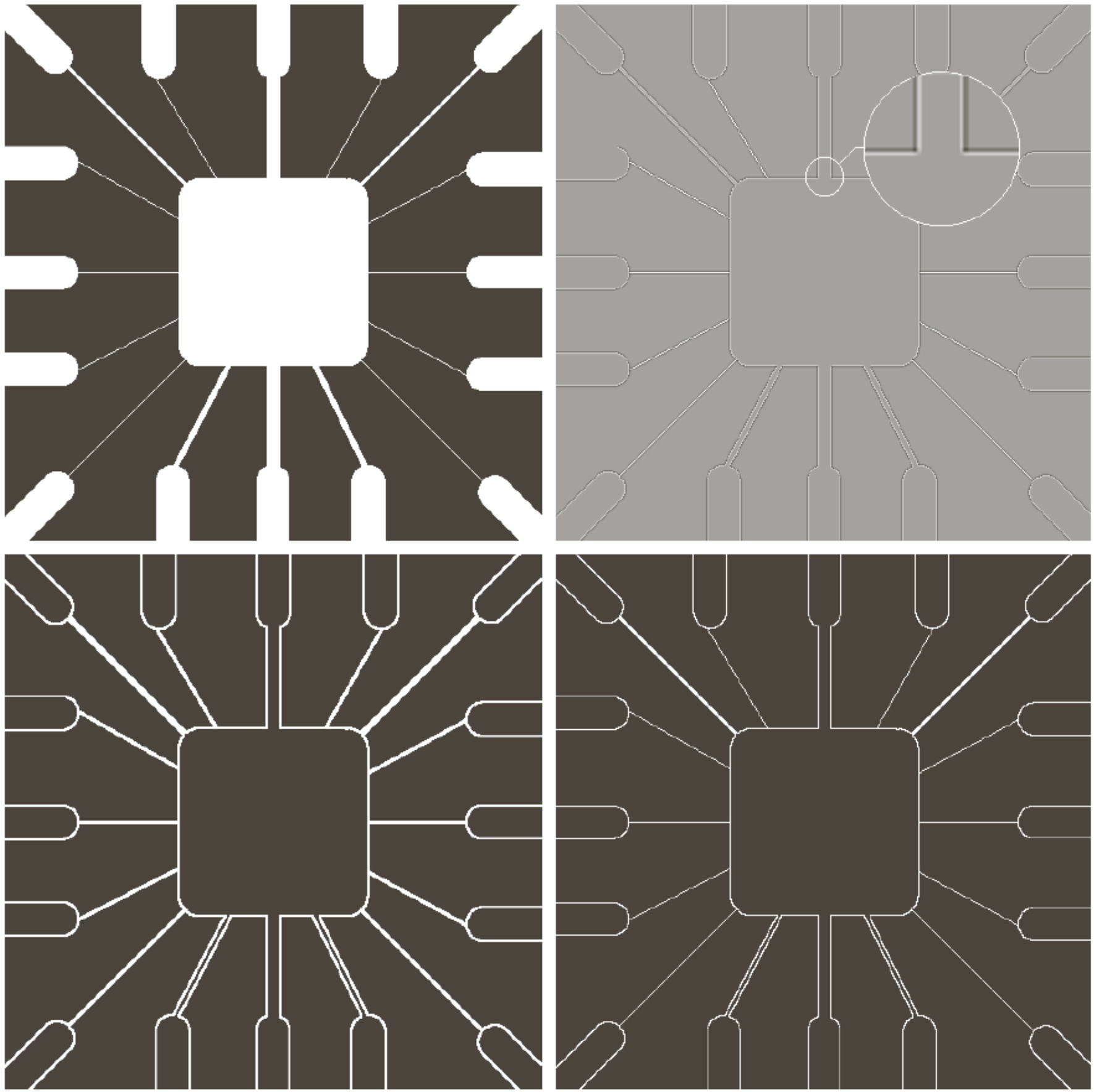
a
b c d

FIGURE 10.4
(a) Point detection (Laplacian) mask. (b) X-ray image of turbine blade with a porosity. The porosity contains a single black pixel. (c) Result of convolving the mask with the image. (d) Result of using Eq. (10.2-8) showing a single point (the point was enlarged to make it easier to see). (Original image courtesy of X-TEK Systems, Ltd.)

Line Detection

Line Detection

- We can use the Laplacian also for detection of line, since it is sensitive to sudden changes and thin lines.
- We must note that since the second derivative changes its sign on a line it creates a “double line effect” and it must be handled.
- Second derivative can have negative results and we need to scale the results



a b
c d

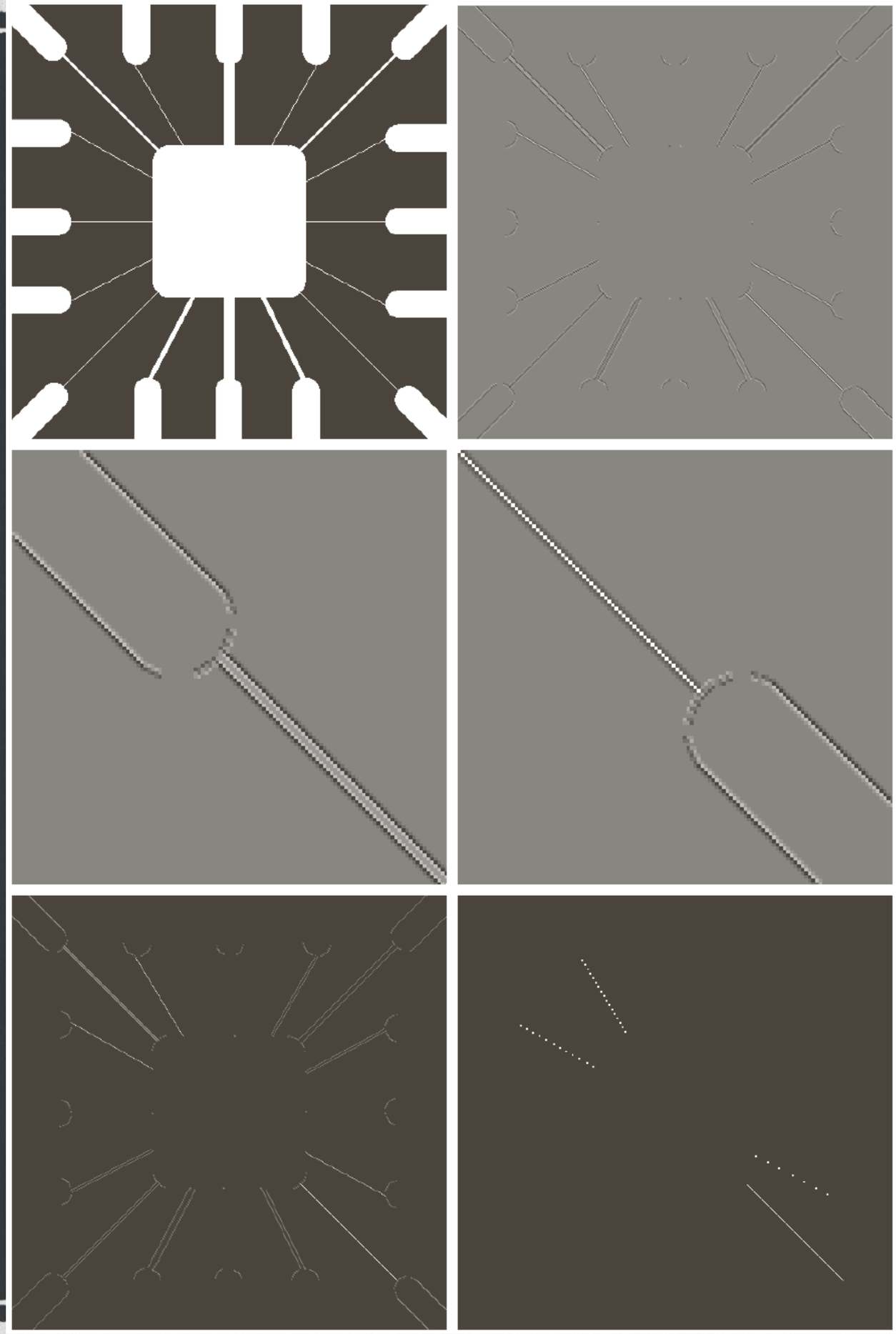
FIGURE 10.5
(a) Original image. (b) Laplacian image; the magnified section shows the positive/negative double-line effect characteristic of the Laplacian. (c) Absolute value of the Laplacian. (d) Positive values of the Laplacian.

Orientation

- The Laplacian is isotropic, i.e. independent of direction.
- If we would like to detect lines on a certain direction only we might want to use masks that would emphasize a certain direction and be less sensitive to other directions.

−1	−1	−1	2	−1	−1	−1	2	−1	−1	−1	2
2	2	2	−1	2	−1	−1	2	−1	−1	2	−1
−1	−1	−1	−1	−1	2	−1	2	−1	2	−1	−1
Horizontal			+45°			Vertical			−45°		

FIGURE 10.6 Line detection masks. Angles are with respect to the axis system in Fig. 2.18(b).



a b
c d
e f

FIGURE 10.7
(a) Image of a wire-bond template. (b) Result of processing with the +45° line detector mask in Fig. 10.6. (c) Zoomed view of the top left region of (b). (d) Zoomed view of the bottom right region of (b). (e) The image in (b) with all negative values set to zero. (f) All points (in white) whose values satisfied the condition $g \geq T$, where g is the image in (e). (The points in (f) were enlarged to make them easier to see.)

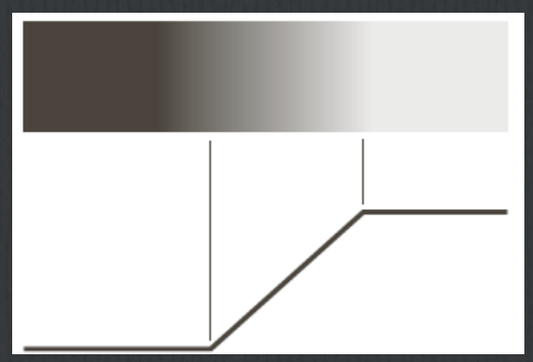
Edge Models

Edge Models

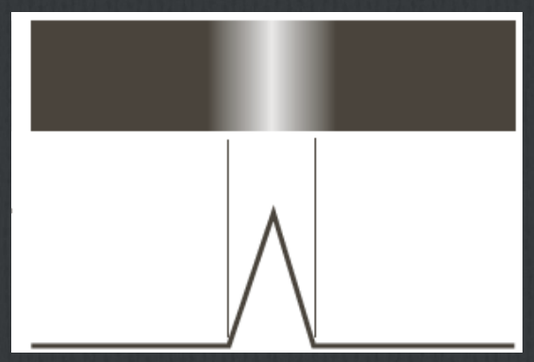
- Edge models are classified according to their intensity profiles:



Step edge – Transition of intensity level over 1 pixel only in ideal, or few pixels on a more practical use



Ramp edge – A slow and graduate transition



Roof edge – A transition to a different intensity and back. Some kind of spread line.

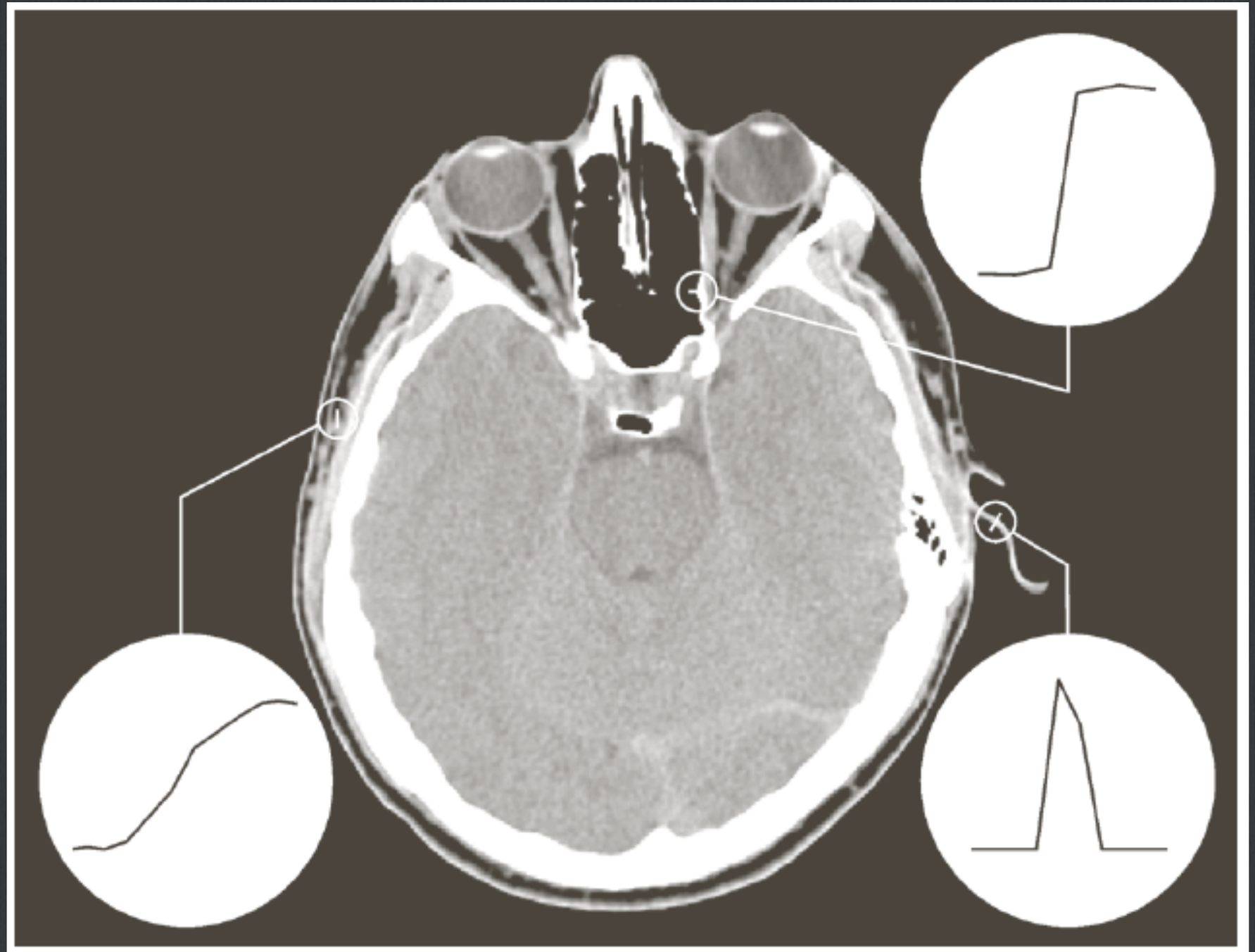
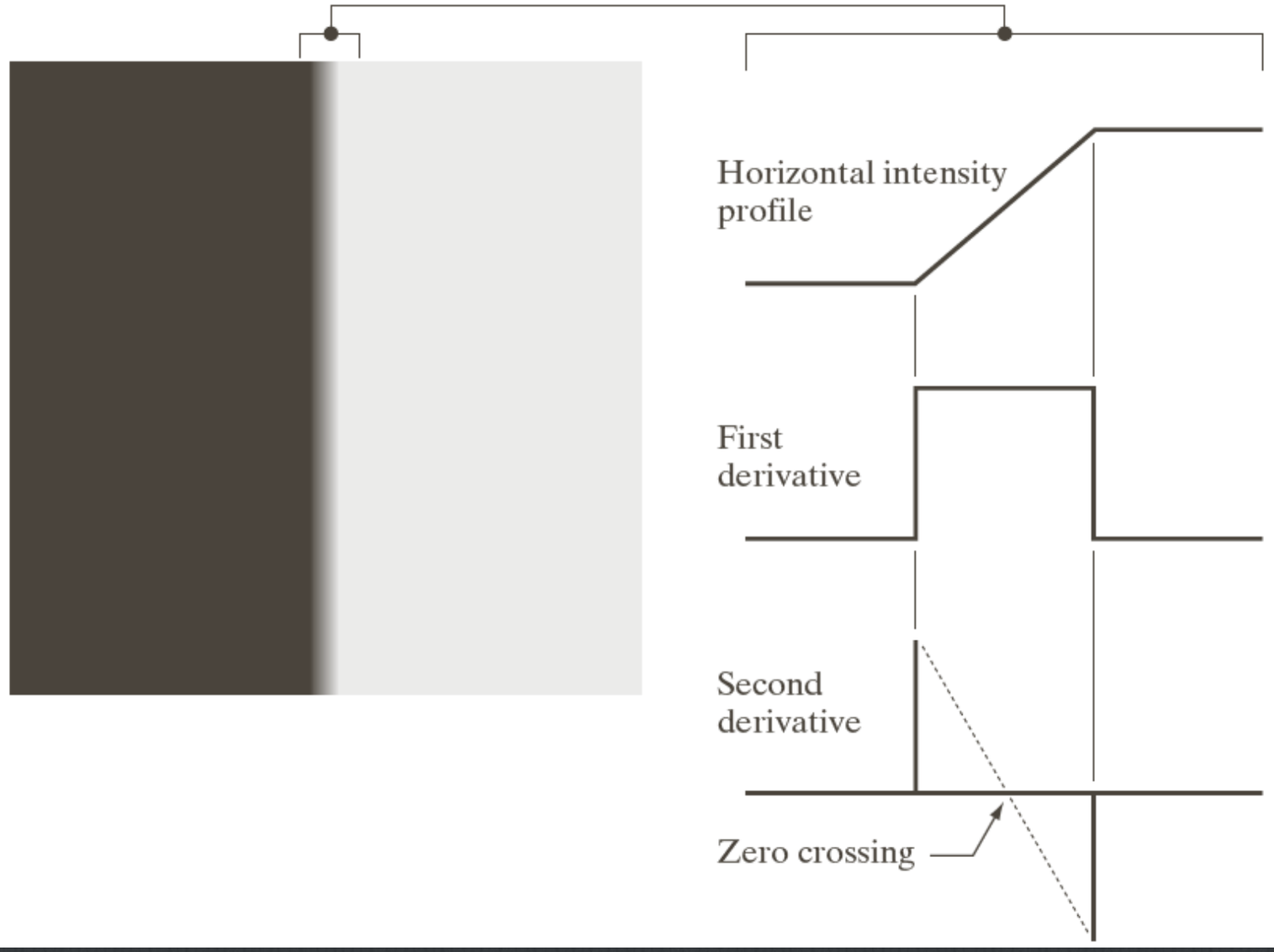


FIGURE 10.9 A 1508 × 1970 image showing (zoomed) actual ramp (bottom, left), step (top, right), and roof edge profiles. The profiles are from dark to light, in the areas indicated by the short line segments shown in the small circles. The ramp and "step" profiles span 9 pixels and 2 pixels, respectively. The base of the roof edge is 3 pixels. (Original image courtesy of Dr. David R. Pickens, Vanderbilt University.)



a b

FIGURE 10.10
(a) Two regions of constant intensity separated by an ideal vertical ramp edge.
(b) Detail near the edge, showing a horizontal intensity profile, together with its first and second derivatives.

- Magnitude of the first derivative can be used to detect an edge
- The sign of the second derivative indicates its direction (black to white or white to black)

- In summary, the three steps performed typically for edge detection are:

1. **Image smoothing for noise reduction.** The need for this step is illustrated by the results in the second and third columns of Fig. 10.11

2. **Detection of edge points.** This is a local operation that extracts from an image all points that are potential edge-point candidates.

3. **Edge localisation.** The objective of this step is to select from the candidate points only the points that are members of the set of points comprising an edge.

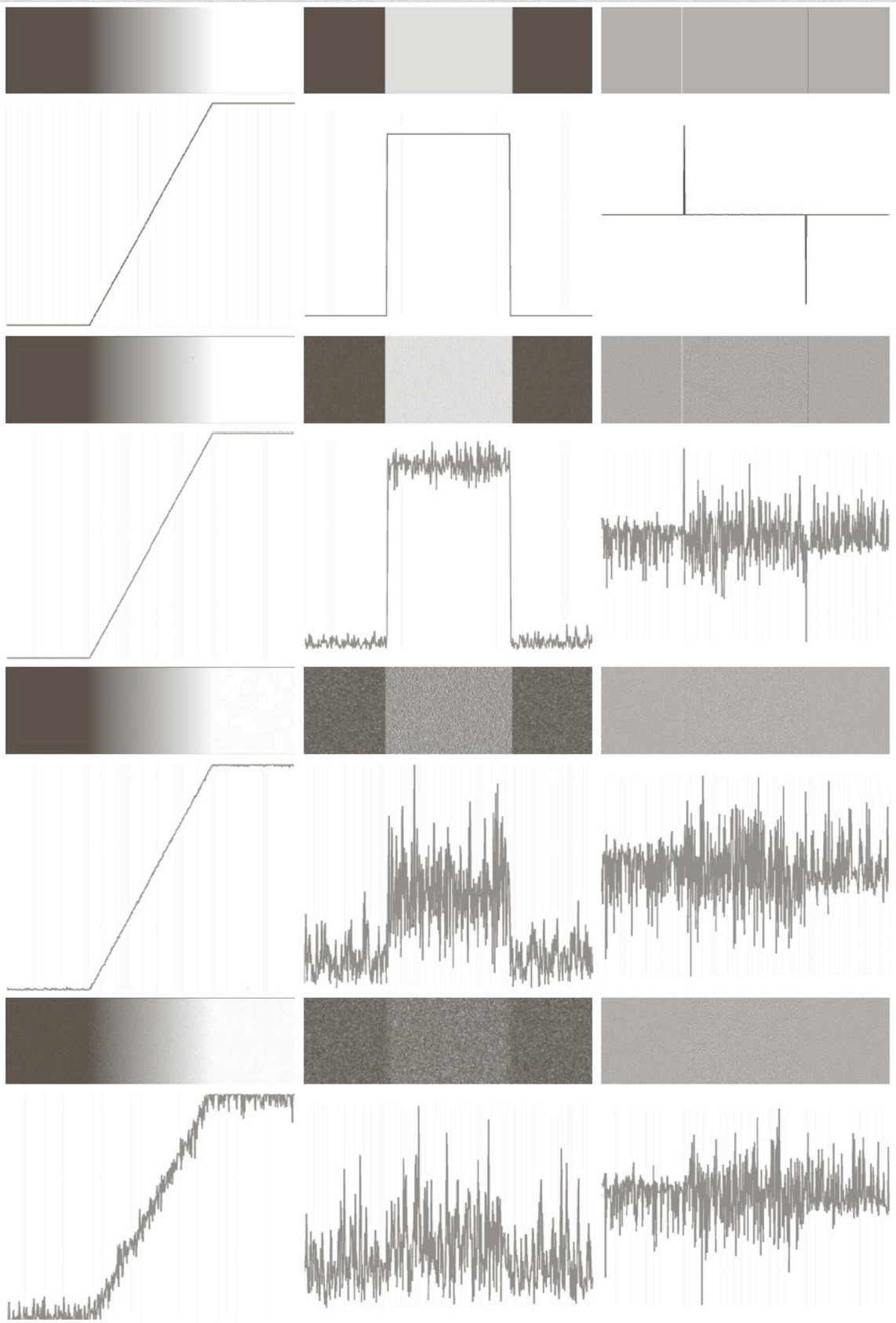


FIGURE 10.11 First column: Images and intensity profiles of a ramp edge corrupted by random Gaussian noise of zero mean and standard deviations of 0.0, 0.1, 1.0, and 10.0 intensity levels, respectively. Second column: First-derivative images and intensity profiles. Third column: Second-derivative images and intensity profiles.

Basic Edge Detection

Image Gradient

- The tool of choice for finding edge strength and direction at location (x,y) of an image, f, is the gradient

$$\nabla f \equiv grad(f) \equiv \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

- The vector has the important geometrical property that it points in the direction of the greatest rate of change of f at location (x,y).

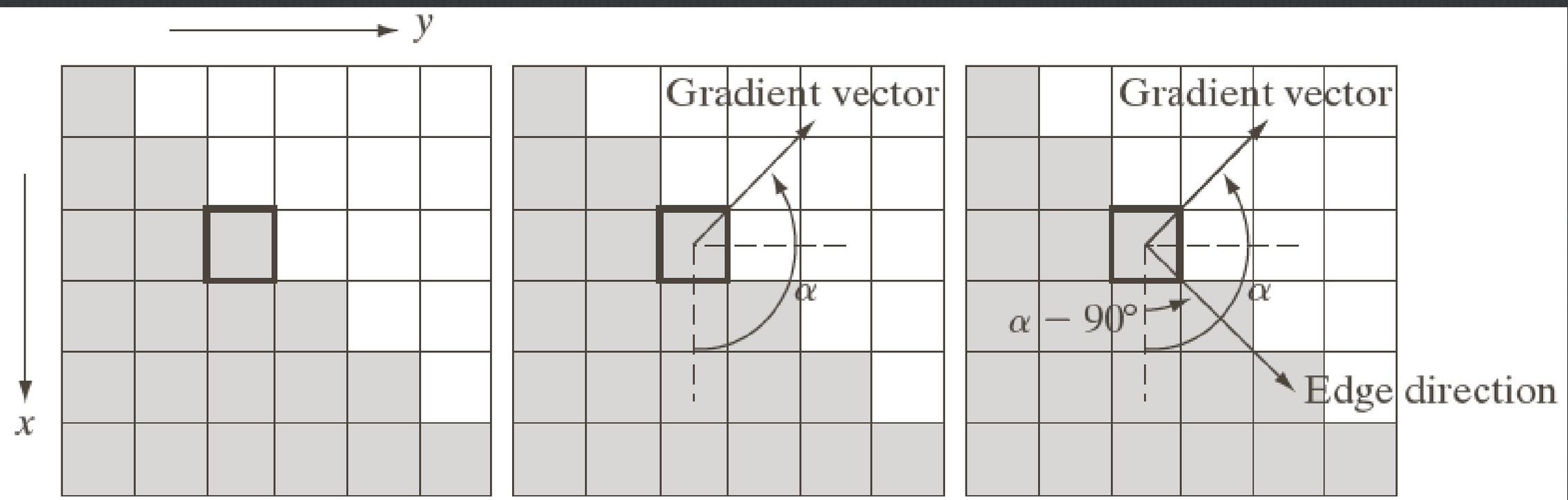
Gradient Properties

- The magnitude of the gradient is the value of the rate of change in the direction of the gradient vector.

$$M(x,y) = mag(\nabla f) = \sqrt{g_x^2 + g_y^2}$$

- The direction of the gradient with respect to the x- axis:

$$\alpha(x,y) = \tan^{-1}\left[\frac{g_y}{g_x}\right]$$



a b c

FIGURE 10.12 Using the gradient to determine edge strength and direction at a point. Note that the edge is perpendicular to the direction of the gradient vector at the point where the gradient is computed. Each square in the figure represents one pixel.

Gradient Operators

- Instead of computing the partial derivatives at every pixel location in the image, we will use approximation.

$$g_x = \frac{\partial f}{\partial x} = f(x+1,y) - f(x,y)$$

$$g_y = \frac{\partial f}{\partial y} = f(x,y+1) - f(x,y)$$

−1
1

−1	1

a b

FIGURE 10.13 One-dimensional masks used to implement Eqs. (10.2-12) and (10.2-13).

- For diagonal edge direction we will use a 2-D mask. Consider the 3x3 region where Z's are intensity values and we want to compute Z_5 gradient.

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

- Mask of 2x2 are simple but they are not as useful for computing edge direction as masks that are symmetric about the center point. Carry more information regarding the direction of an edge.

Prewitt Operators

Rows subtract x-direction

$$g_x = \frac{\partial f}{\partial x} = (z_7 + z_8 + z_9) - (z_1 + z_2 + z_3)$$

−1	−1	−1
0	0	0
1	1	1

Columns subtract y-direction

$$g_y = \frac{\partial f}{\partial y} = (z_3 + z_6 + z_9) - (z_1 + z_4 + z_7)$$

−1	0	1
−1	0	1
−1	0	1

Sobel Operators

- Using a 2 in the center location provides image smoothing

$$g_x = \frac{\partial f}{\partial x} = (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)$$

−1	−2	−1
0	0	0
1	2	1

$$g_y = \frac{\partial f}{\partial y} = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)$$

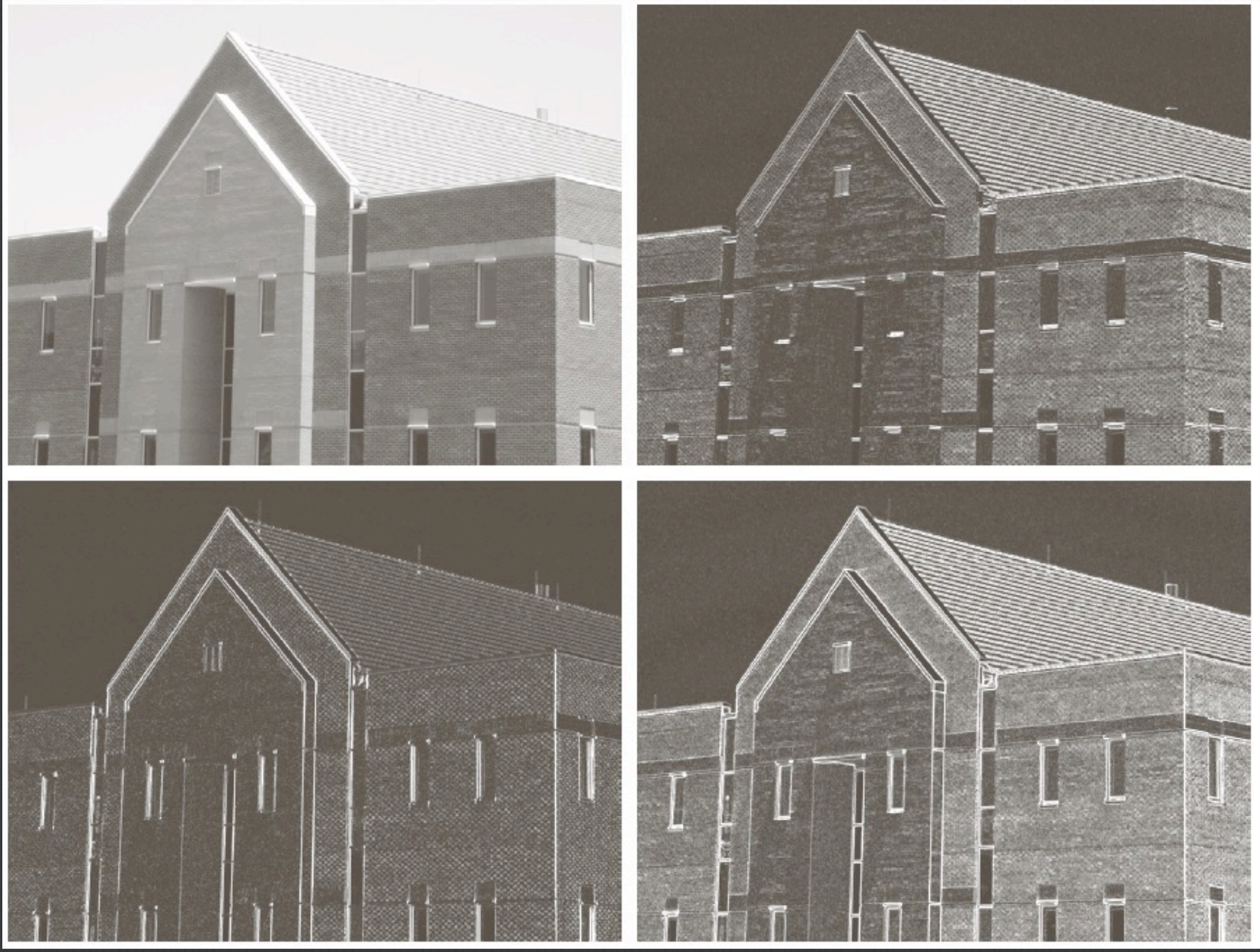
−1	0	1
−2	0	2
−1	0	1

Prewitt	Sobel
Simpler to implement	Better noise suppression (smoothing)
Give isotropic results only for vertical and horizontal edges	Give isotropic results only for vertical and horizontal edges
	Preferred because noise suppression is an important issue when dealing with derivatives .

Computing the Magnitude

- Requires that g_x and g_y be combined. However, it is hard to compute thus we will use approximate value.

$$M(x,y) \approx |g_x| + |g_y|$$



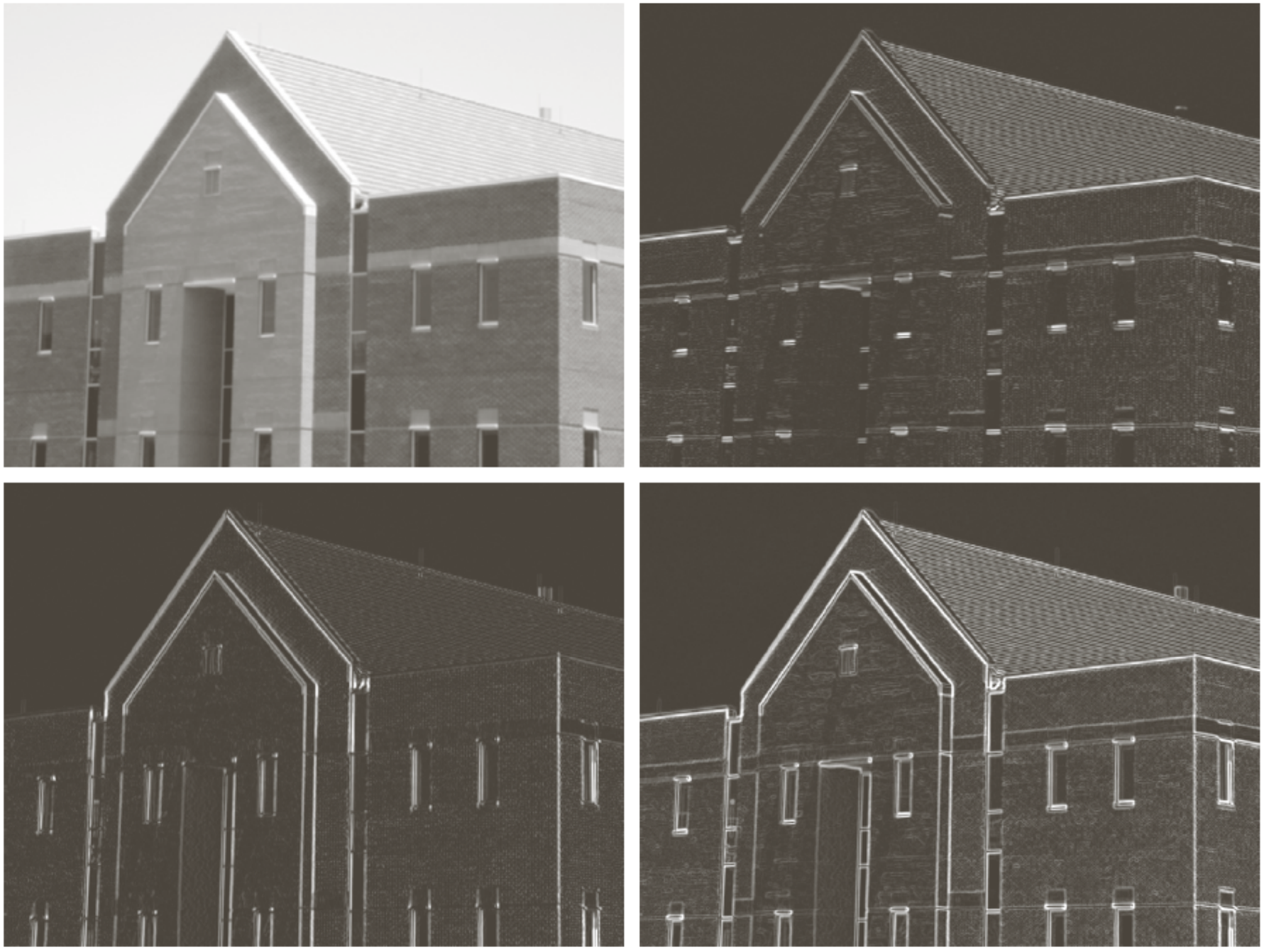
a b
c d

FIGURE 10.16
(a) Original image of size 834 × 1114 pixels, with intensity values scaled to the range [0, 1]. (b) $|g_x|$, the component of the gradient in the x-direction, obtained using the Sobel mask in Fig. 10.14(f) to filter the image. (c) $|g_y|$, obtained using the mask in Fig. 10.14(g). (d) The gradient image, $|g_x| + |g_y|$.

Combining the Gradient with Thresholding

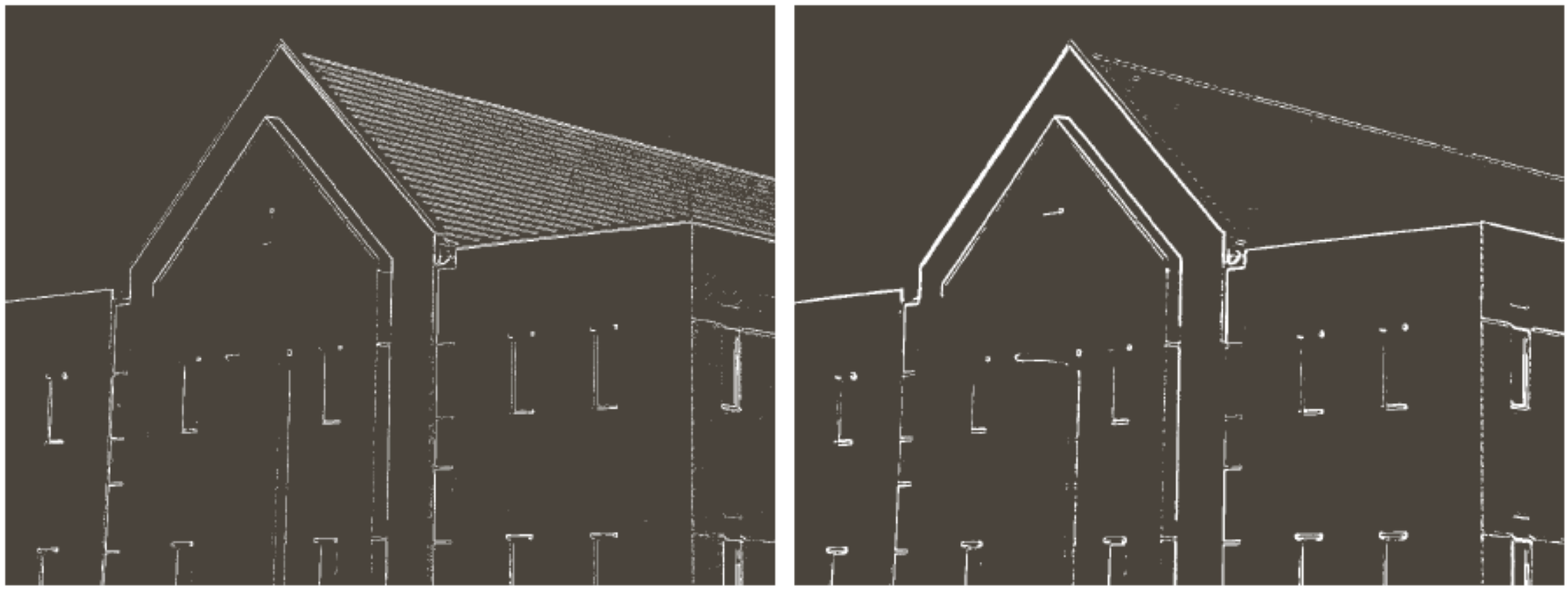
- Sometimes we would use threshold on the gradient image to achieve the same results.

- Pixels with values greater than threshold are shown white and the other are shown black.



a b
c d

FIGURE 10.18
Same sequence as in Fig. 10.16, but with the original image smoothed using a 5×5 averaging filter prior to edge detection.



a b

FIGURE 10.20 (a) Thresholded version of the image in Fig. 10.16(d), with the threshold selected as 33% of the highest value in the image; this threshold was just high enough to eliminate most of the brick edges in the gradient image. (b) Thresholded version of the image in Fig. 10.18(d), obtained using a threshold equal to 33% of the highest value in that image.